

5-DOF Robotic Arm

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1 Introduction

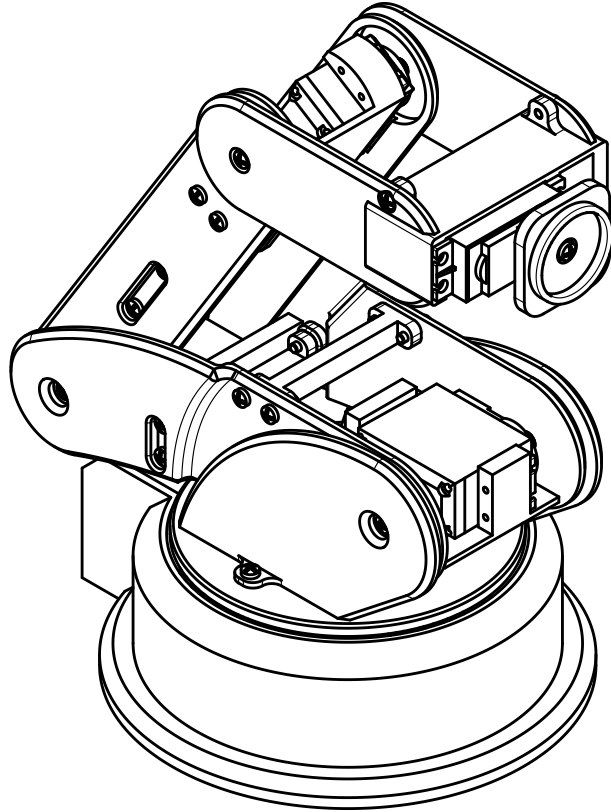


Figure 1: Full arm assembly

This project is a simple yet modular and easily modifiable 5-DOF robotic arm intended to be a platform for the teaching and learning of basic robotics control systems. The project uses five MG996R servo motors as they are a common hobbyist robotics component. Figure 1 shows the final assembled arm in a resting position.

The arm does not possess an actuated gripper as it has been designed with each servo placed directly within the arm's joints in order to avoid the complexities of gearboxes or belt drives. This allows for the stated focus on control systems and means that any issues encountered in actuation of the arm can be immediately attributed to the servos themselves, streamlining the physical troubleshooting process.

2 Drawings

The arm is comprised of six assemblies. As can be seen in Figure 2, this includes the base of the arm, four link pieces, and a plate at the end of the arm to visualise actuation of the final servo.

3 Bill of Materials (BOM)

4 Calculations

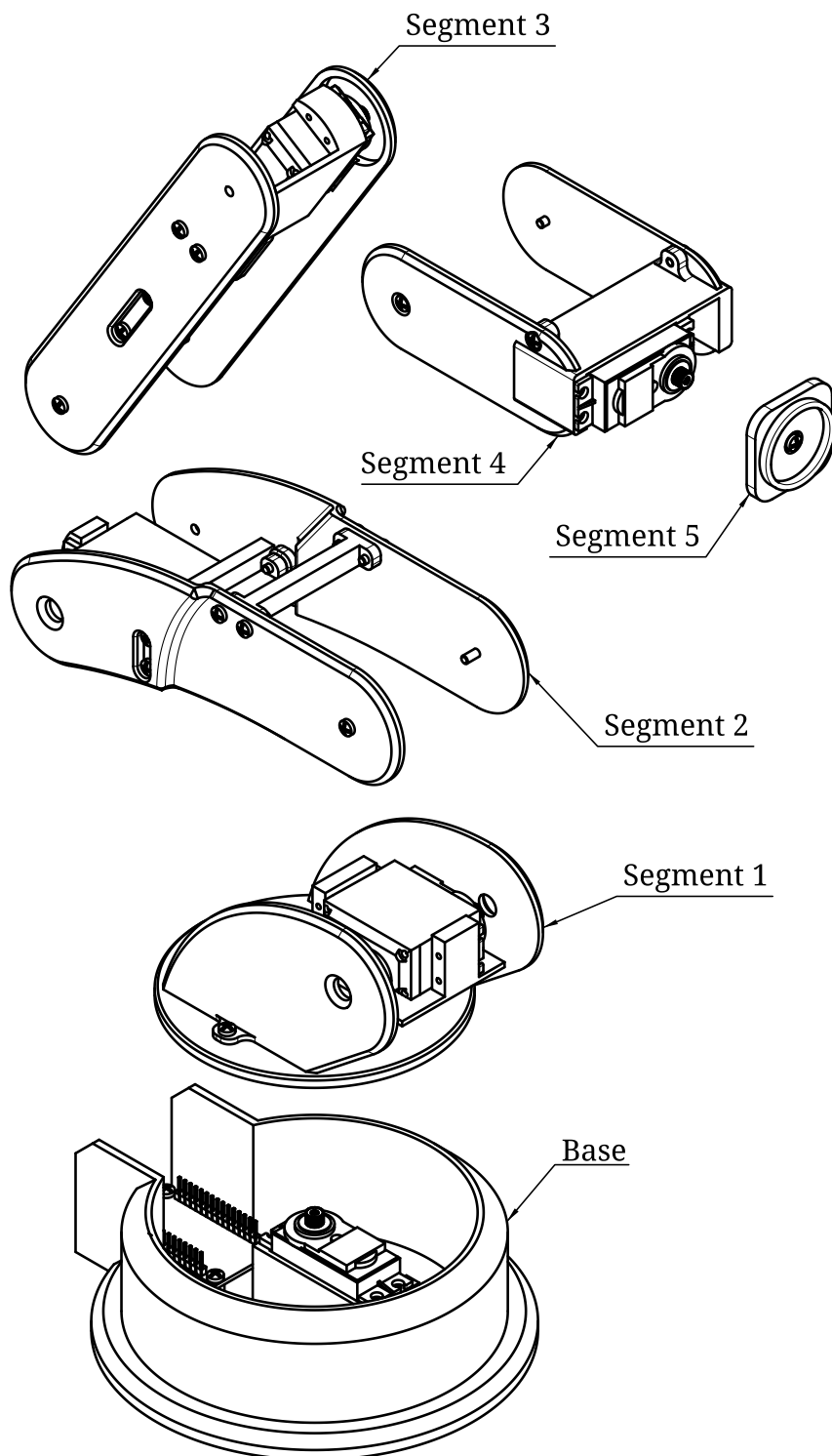


Figure 2: Exploded arm assembly

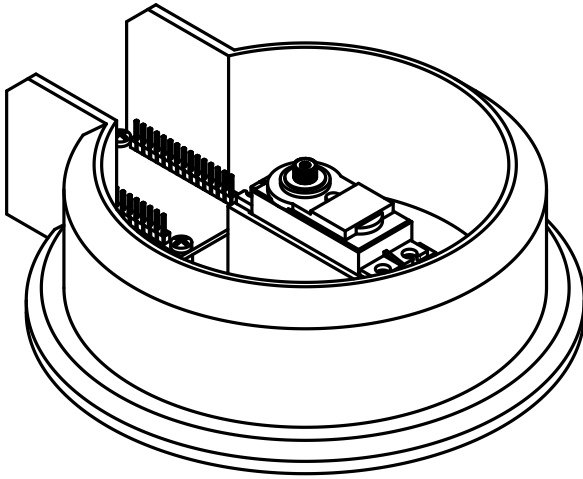


Figure 3: Base assembly

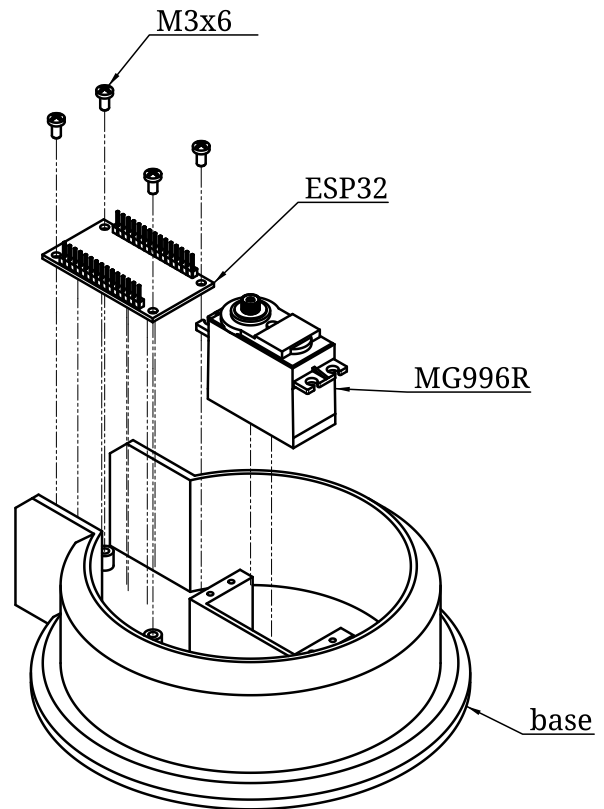


Figure 4: Base exploded view

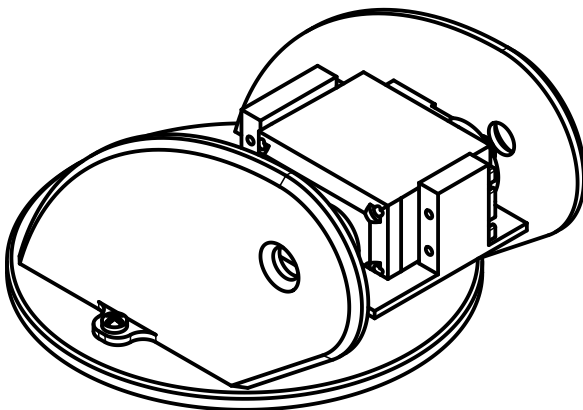


Figure 5: Segment 1 assembly

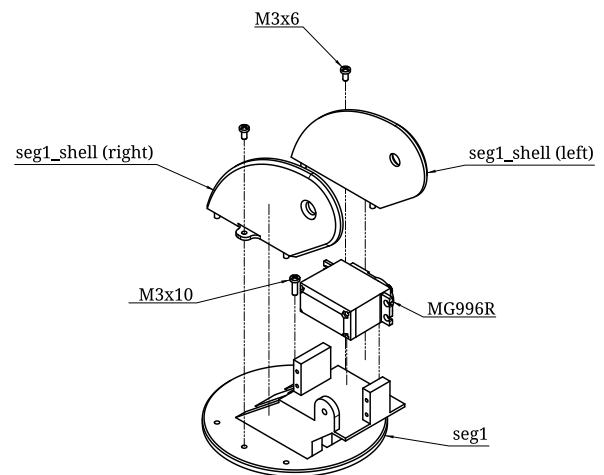


Figure 6: Segment 1 exploded view

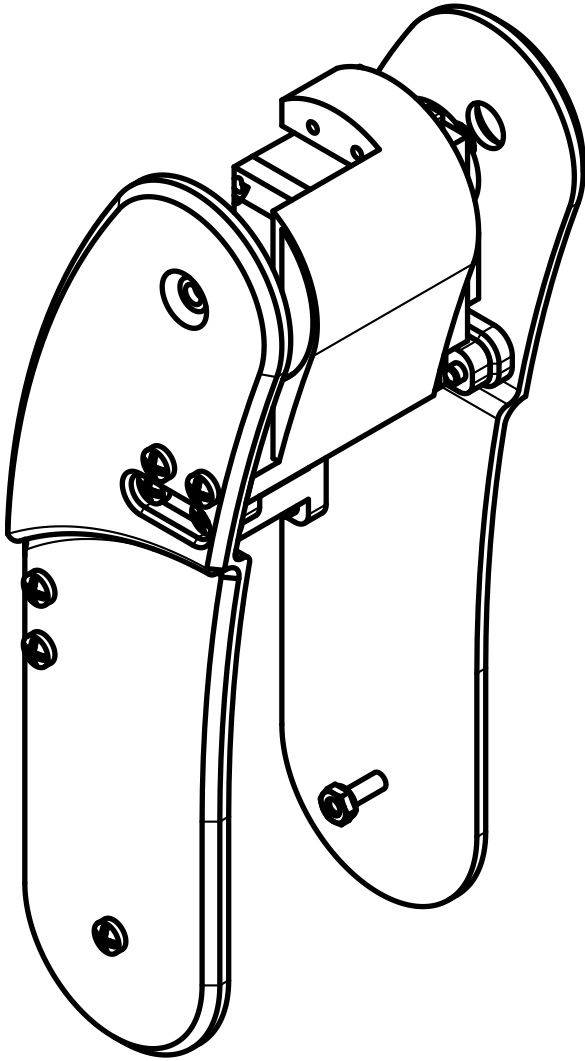


Figure 7: Segment 2 assembly

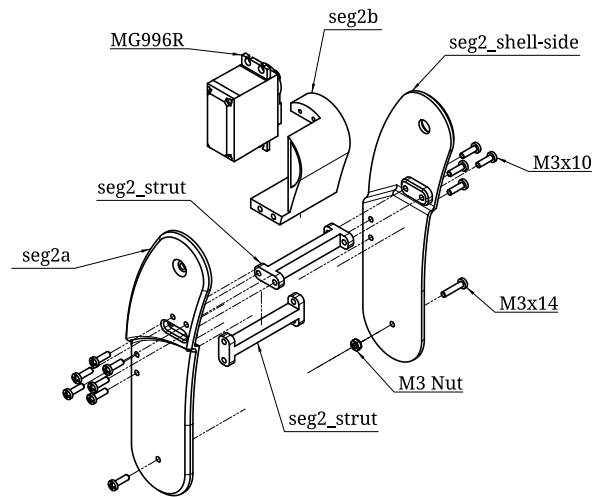


Figure 8: Segment 2 exploded view

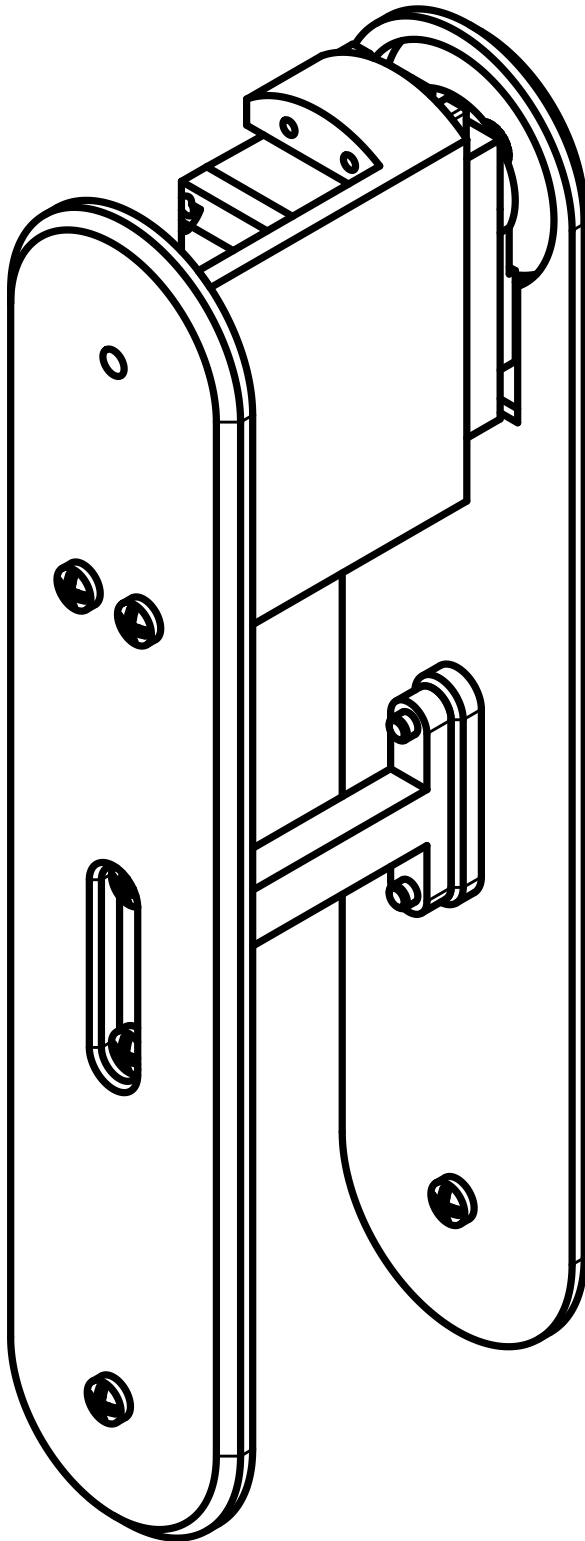


Figure 9: Segment 3 assembly

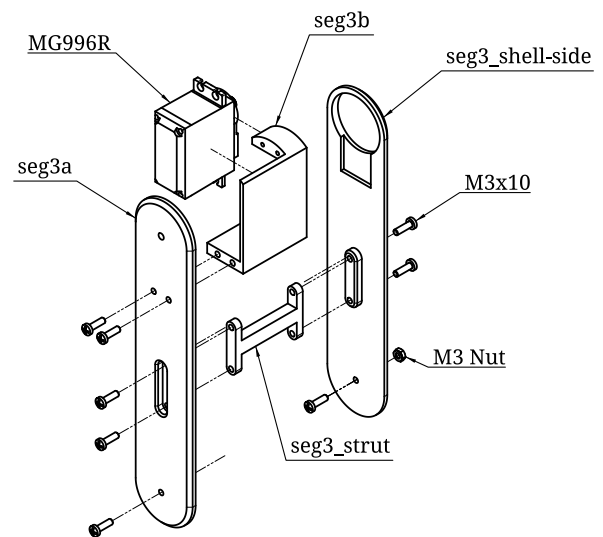


Figure 10: Segment 3 exploded view

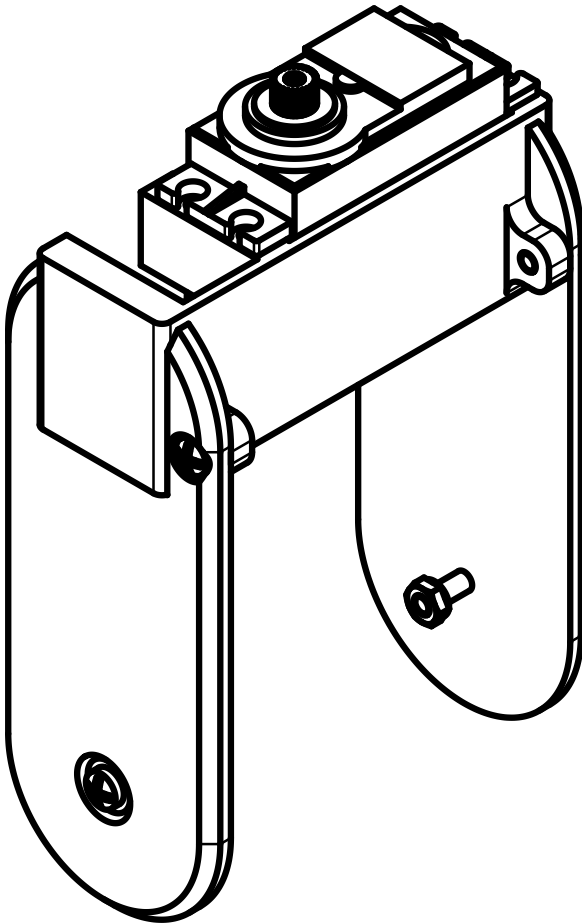


Figure 11: Segment 4 assembly

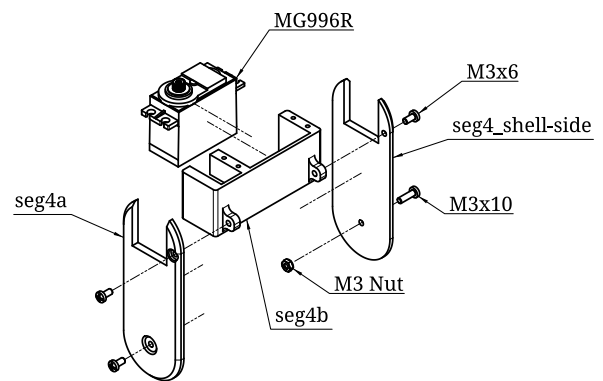


Figure 12: Segment 4 exploded view

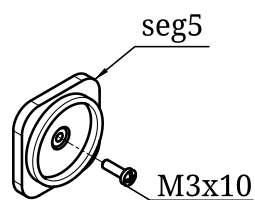


Figure 13: Exploded arm assembly